Module 1: Critical Thinking

Start Assignment

- Points 50
- Submitting a file upload

Critical Thinking Assignment (50 Points)

For this assignment, you will write a simple Python prototype and then create a short developer report, simulating the kind of deliverable expected by a technical team lead or project manager in a modern software development environment. Whether you're working independently or as part of a team, understanding the challenges of software development and communicating technical work clearly are essential skills for success.

To get familiar with Python programming and integrated development tools used in the field, begin by writing a Python script with a practical, real-world purpose. Then, create a brief developer-facing report reflecting on key aspects of your experience and how they relate to broader software development challenges.

Part 1: Python Script Development

Write a Python script that performs a useful or realistic task. You may choose one of the following or propose your own idea:

- A command-line tip calculator.
- A script that reads and analyzes a CSV file (e.g., grades, sales).
- A simple to-do list or task tracker.
- A basic simulation (e.g., mock CPU/memory monitor using random values).
- A log parser that searches for keywords in a sample file.

Use any modern IDE of your choice (e.g., VS Code, PyCharm, Replit).

Part 2: Developer Report (Professional Documentation)

After completing your script, prepare a developer report answering the following stakeholder-facing questions based on your experience:

- What was the purpose or intended use case of your script?
- What tools or libraries did you use, and why?
- What challenges did you encounter during development?
- How would you expand or improve this prototype in future iterations?
- What lessons did you learn that apply to broader software development work?

Keep your responses brief and professional (1–3 sentences per question). This report simulates what you might send as part of an internal project handoff or sprint demo.

Part 3: Submission Requirements

Submit the following in a zipped folder:

- Your Python source file (.py).
- A screenshot of your program running successfully.
- Your completed developer report (.txt, .md, or .pdf).

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Criteria	Ratings				Pts
Part 1: Python Script Development - Functionality & Usefulness	**15 to >12.0 pts** **Meets Expectation** The Python script fully implements its stated purpose, performing a useful and realistic task effectively. All features work as intended without errors.	**12 to >9.0 pts** **Approaches Expectation** The Python script largely implements its stated purpose, performing a mostly useful task. Most features work, but there may be minor bugs or limited functionality.	**9 to >6.0 pts** **Below Expectation** The Python script implements some features but is incomplete, contains significant errors, or its purpose is not clearly useful/realistic.	**6 to >0 pts** **Limited Evidence** The Python script is minimal, non-functional, or irrelevant to the assignment's purpose.	15 pts
Part 1: Python Script Development - Code Quality & Readability	**10 to >8.0 pts** **Meets Expectation** The Python code is exceptionally well-structured, clear, readable, and includes appropriate comments. It demonstrates good programming practices and adheres to basic Pythonic conventions.	**8 to >6.0 pts** **Approaches Expectation** The Python code is generally well-structured and readable, with some comments. It demonstrates acceptable programming practices, though some improvements could be made.	**6 to >4.0 pts** **Below Expectation** The Python code is poorly structured, difficult to read, or lacks essential comments. It exhibits significant programming flaws or inconsistencies.	**4 to >0 pts** **Limited Evidence** The Python code is unreadable, disorganized, or demonstrates a severe lack of programming understanding.	10 pts
Part 2: Developer Report - Content & Completeness	**15 to >12.0 pts** **Meets Expectation** The developer report comprehensively and insightfully answers all five stakeholder-facing questions. Responses are highly relevant, demonstrate strong reflection on the development process, and show understanding of broader software development challenges.	**12 to >9.0 pts** **Approaches Expectation** The developer report answers all five questions, but some responses may lack depth, insight, or full relevance to the development experience or broader challenges.	**9 to >6.0 pts** **Below Expectation** The developer report is incomplete, missing answers to some questions, or responses are superficial, vague, or irrelevant to the prompts.	**6 to >0 pts** **Limited Evidence** The developer report is largely absent, or responses are minimal, inaccurate, or unrelated to the assignment.	15 pts
Part 2: Developer Report - Professionalism & Conciseness	**5 to >4.0 pts** **Meets Expectation** The developer report is highly professional in tone and language. All responses strictly adhere to the 1-3 sentence limit, conveying information clearly and concisely for a technical audience.	**4 to >3.0 pts** **Approaches Expectation** report is highly professional in tone and language. All responses strictly adhere to the 1-3 sentence limit, conveying information clearly and concisely for a technical audience. The developer report is generally professional in tone. Most responses adhere to the 1-3 sentence limit, but some may be slightly too long or less concise.	**3 to >2.0 pts** **Below Expectation** The developer report lacks professionalism in tone or language. Responses frequently exceed the 1-3 sentence limit, or clarity is significantly compromised.	**2 to >0 pts** **Limited Evidence** The developer report is unprofessional, unreadable, or fails to address conciseness requirements.	5 pts
Part 3: Submission Requirements & Successful Execution	**5 to >4.0 pts** **Meets Expectation** All required submission files (.py, screenshot, .txt/.md/.pdf report) are correctly included in a single zipped folder. The screenshot clearly demonstrates the program running successfully with its intended output/functionality.	**4 to >3.0 pts** **Approaches Expectation** Most required submission files are included and correctly formatted. The screenshot is present and generally shows the program running, though it might lack optimal clarity.	**3 to >2.0 pts** **Below Expectation** Missing one or more required submission files, or files are incorrectly formatted/not zipped. The screenshot is unclear, or the program does not run successfully.	**2 to >0 pts** **Limited Evidence** Major submission components are missing, incorrect, or the program cannot be executed.	5 pts
					Total Points: 50